



# CIVITHON

2026



CIVITHON is a student-focused hackathon organized by the Civitronicz Club. It aims to bring together creative minds to solve real-world problems related to infrastructure, environment, and smart city development. Participants will work in teams to design innovative and practical solutions using modern technologies.

## RULES

- Participants must register in teams (2–4 members).
- Each team must choose one problem statement from the provided list.
- Use of any programming language, tools, or platforms is allowed.
- Pre-built projects are not allowed; work must be done during the hackathon.
- Evaluation will be based on innovation, feasibility, technical implementation, and presentation.
- Plagiarism or copying will lead to disqualification.
- Judges' decision will be final.

## PROBLEM STATEMENTS WITH EXPLANATION

### 1. Flood Prediction and Smart Drainage System

Urban flooding causes major damage due to poor drainage and lack of monitoring. A smart system can use sensors and weather data to predict floods and manage water flow effectively.

### 2. Coastal Erosion Monitoring and Protection System

Coastal areas are losing land due to erosion and climate changes. A system using sensors or data analysis can help monitor erosion and provide solutions to protect these regions.

### 3. Traffic Management During Metro Construction in Bengaluru

Metro construction leads to heavy traffic congestion, delays, and pollution. A smart traffic system can use real-time data and AI to manage traffic and suggest better routes.

### 4. Smart Monitoring and Preservation of Aging Structures

Old buildings and bridges may become unsafe over time. Smart sensors can be used to detect cracks and structural issues early, helping prevent accidents.

### 5. Smart Urban Heat Island Reduction System

Cities become hotter due to concrete and lack of greenery. Smart solutions like green roofs and reflective materials can help reduce temperature and improve comfort.

### 6. Smart Infrastructure Health Monitoring System

Continuous monitoring of infrastructure is important for safety. IoT-based systems can track the condition of roads and structures and provide alerts for maintenance.

### 7. Smart Parking and Urban Space Optimization

Parking issues increase traffic and waste time. A smart parking system can guide users to available spaces and improve urban space usage.

### 8. IoT and Data Analysis to Eliminate Non-Revenue Water

Water loss due to leaks is a major issue. IoT sensors and data analysis can help detect leaks early and reduce wastage.



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## ***TEMPLATE***



***Project Title*** \_\_\_\_\_

Team Name

Team Members: Name 1 | Name 2 | Name 3

Problem Statement Chosen





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## ***PROBLEM OVERVIEW***

- What is the problem?
- Where does it occur?
- Why is it important?
- Who is affected?



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## ***SOLUTION OVERVIEW***

- Our solution in one line: \_\_\_\_\_
- Key idea: \_\_\_\_\_
- What makes it innovative: \_\_\_\_\_



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## ***SOLUTION STACK***

- Technologies used: \_\_\_\_\_
- Tools / Platforms: \_\_\_\_\_
- Hardware (if any): \_\_\_\_\_



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## ***APPROACH***

- Step 1: \_\_\_\_\_
- Step 2: \_\_\_\_\_
- Step 3: \_\_\_\_\_
- (Add diagram/flowchart here)



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## ***IMPACT***

- Target users: \_\_\_\_\_
- Social impact: \_\_\_\_\_
- Environmental impact: \_\_\_\_\_
- Economic benefits: \_\_\_\_\_





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## ***FEASIBILITY***

- Technical feasibility: \_\_\_\_\_
- Scalability: \_\_\_\_\_
- Sustainability: \_\_\_\_\_
- Real-world viability: \_\_\_\_\_





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## ***PROTOTYPE / MODEL***

- Description of model/demo
- Screenshots / images
- Working explanation



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## ***CONCLUSION***

- Summary of solution
- Key highlights
- Why this should be implemented



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***THANK YOU***

Thank You!

(Optional: QR Code for demo / GitHub / video)